Contents

1 Terminology, the Standard Human, and Scaling 1
1.1 Anatomical Terminology 2
1.2 Motion in the Human Machine 5
1.3 The Standard Human 16
1.4 Scaling Relationships 23
1.4.1 Allometric Rules 24
1.4.2 Scaling in the Senses 26
1.5 Summary 27
Problems 27
References 37

2 Statics of the Body 39
2.1 Review of Forces, Torques, and Equilibrium 39
2.2 Statics: Motion in One Plane and Levers 42
2.3 Statics in the Body 46
2.3.1 The Lower Arm 46
2.3.2 Hip Problems 52
2.3.3 Statics of Other Synovial Joints 61
2.3.4 Lower Back Problems 69
2.3.5 Three-Force Rule 78
2.3.6 Multisegment Modeling 79
2.4 The Sense of Touch 81
2.5 Diversion into the Units of Force and Pressure 82
2.5.1 Force 82
2.5.2 Pressure 83
2.6 Summary 84
Problems 85
References 93

3 Motion 95
3.1 Kinematics and Musculature 98
3.2 Standing 105

3.2.1 Stability . . . 105
3.2.2 Forces on the Feet . . . 109
3.3 Walking . . . 109
3.3.1 Kinematics . . . 110
3.3.2 Muscular Action . . . 112
3.3.3 Friction . . . 113
3.3.4 Energetics . . . 118
3.3.5 Review of Harmonic Motion, Pendulums, and Moments of Inertia . . . 121
3.3.6 Ballistic (or Pendulum) Model of Walking . . . 126
3.3.7 Inverted Pendulum Model . . . 129
3.4 Running . . . 130
3.4.1 Kinematics . . . 130
3.4.2 Muscular Action . . . 132
3.4.3 Energetics . . . 133
3.4.4 Bouncing Ball/Pogo Stick Model . . . 139
3.5 Jumping . . . 140
3.5.1 Vertical Jump . . . 141
3.5.2 Pole Vault . . . 145
3.5.3 High Jump and Long Jump . . . 147
3.6 Throwing a Ball . . . 150
3.6.1 Comparing Throwing Motions . . . 163
3.6.2 Power Generated During a Throw . . . 164
3.7 Batting and Swinging Motions . . . 165
3.8 Other Types of Motion . . . 167
3.9 Multisegment Modeling (Advanced Topic) . . . 170
3.10 Collisions of the Human Body . . . 173
3.10.1 Kinematics of a Collision . . . 173
3.10.2 Consequences of Collisions . . . 175
3.10.3 Contact Sports . . . 182
3.10.4 Hitting Balls . . . 187
3.10.5 Running . . . 191
3.10.6 Jumping . . . 193
3.10.7 Chewing Food . . . 193
3.11 Sustained Acceleration . . . 194
3.12 Flight of the Ball . . . 195
3.12.1 The Effect of Air Drag on Moving Balls and Humans . . . 196
3.12.2 The Flight of a Spinning Ball . . . 202
3.12.3 Bouncing, Rolling and Hitting Balls, including those with Spin . . . 204
3.13 Summary . . . 210
Problems . . . 210
References . . . 240

4 Mechanical Properties of the Body 247
4.1 Material Components of the Body 250
4.1.1 Bone 251
4.1.2 Ligaments and Tendons 253
4.1.3 Cartilage 254
4.2 Elastic Properties 255
4.2.1 Basic Stress–Strain Relationships 255
4.2.2 Other Stress–Strain Relations 258
4.2.3 Bone Shortening 259
4.2.4 Energy Storage in Elastic Media 260
4.3 Time-Independent Deviations in Hookean Materials 263
4.3.1 Non-Hookean Materials 272
4.4 Static Equilibrium of Deformable Bodies (Advanced Topic) 274
4.4.1 Bending of a Beam (or Bone) 279
4.5 Time-Dependent Deviations from Elastic Behavior: Viscoelasticity 283
4.5.1 Perfect Spring 287
4.5.2 Perfect Dashpot 287
4.5.3 Simple Viscoelastic Models 289
4.6 Viscoelasticity in Bone 297
4.7 Bone Fractures 299
4.7.1 Modes of Sudden Breaking of Bones 300
4.7.2 Stress Fractures (Advanced Topic) 307
4.8 Common Sports Injuries 311
4.9 Avoiding Fractures and Other Injuries: Materials for Helmets 314
4.10 Mechanical Properties of Food 317
4.11 Summary 319
Problems 319
References 326

5 Muscles 331
5.1 Skeletal Muscles in the Body 332
5.1.1 Types of Muscle Activity 335
5.2 The Structure of Muscles 336
5.3 Activating Muscles: Macroscopic View 341
5.3.1 Mechanical Model of the Active State of Muscles 344
5.4 Passive Muscles 350
5.5 Active/Tetanized Muscles: Microscopic View 351
5.5.1 Total Muscle Tension 352
5.5.2 Everyday Proof of the Limited Range of Useful Muscle Length 353
5.6 Hill Force–Velocity Curve 355
5.7 The Combined Muscle-Tendon System 362

5.8 The Sliding Filament Model: Nanoscopic View . . . 363
5.8.1 Statistical Model of the Hill Force–Velocity Curve (Advanced Topic) . . . 368
5.9 Titin . . . 370
5.10 Coordination of Muscles . . . 371
5.11 Muscle Strength and Evolution . . . 374
5.11.1 Increasing Strength with Training . . . 374
5.11.2 Muscle Evolution with Age . . . 376
5.11.3 Muscle Fatigue . . . 377
5.12 Summary . . . 378
Problems . . . 378
References . . . 388

6 Metabolism: Energy, Heat, Work, and Power of the Body . . . 393
6.1 Conservation of Energy and Heat Flow . . . 394
6.2 Energy Content of Body Fuel . . . 396
6.2.1 Metabolizable Energy . . . 400
6.2.2 Energy Storage . . . 402
6.3 Energy Storage Molecules . . . 405
6.3.1 How ATP Is Produced and Used as an Energy Source . . . 405
6.3.2 How ATP Is Actually Used by the Body . . . 407
6.4 Metabolic Rates . . . 415
6.4.1 Basal Metabolic Rate . . . 415
6.4.2 Metabolic Rates During Common Activities . . . 421
6.4.3 Weight Gain and Loss . . . 438
6.5 Loss of Body Heat . . . 442
6.5.1 Modes of Heat Loss . . . 444
6.6 Body Temperature . . . 459
6.7 Summary . . . 466
Problems . . . 467
References . . . 485

7 Fluid Pressure, Fluid Flow in the Body, and Motion in Fluids . . . 491
7.1 Characteristic Pressures in the Body . . . 492
7.1.1 Definition and Units . . . 492
7.1.2 Measuring Pressure . . . 492
7.2 Basic Physics of Pressure and Flow of Fluids . . . 494
7.2.1 Law of Laplace . . . 495
7.2.2 Fluids in Motion . . . 497
7.2.3 Equation of Continuity . . . 498
7.2.4 Bernoulli's Equation . . . 499
7.2.5 Interactions Among the Flow Parameters . . . 500
7.2.6 Resistance in Flow: Viscous Flow and Poiseuille's Law . . . 501

7.2.7 Compliance in Flow . . . 509
7.2.8 Inertance in Flow . . . 511
7.2.9 Flow Under Special Conditions . . . 512
7.2.10 Flow of Objects in Fluids: Drag and Lift . . . 513
7.3 Diffusion (Advanced Topic) . . . 515
7.4 Pressure and Flow in the Body . . . 518
7.5 Motion of Humans in Fluids . . . 519
7.5.1 Swimming . . . 520
7.5.2 Human Flight . . . 523
7.6 Summary . . . 524
Problems . . . 525
References . . . 530

8 Cardiovascular System . . . 533
8.1 Overview of the Circulatory System and Cardiac Cycle . . . 533
8.1.1 Circulation . . . 533
8.1.2 Cardiac Cycle . . . 541
8.1.3 Valves . . . 544
8.2 Physics of the Circulation System . . . 546
8.2.1 Properties of Blood . . . 546
8.2.2 Blood Pressure and Flow in Vessels . . . 547
8.2.3 Capillaries and Osmotic Pressure . . . 561
8.2.4 Blood Flow Rates and Speeds . . . 564
8.2.5 Consequences of Clogged Arteries . . . 572
8.2.6 Work Done by the Heart and the Metabolic Needs of the Heart . . . 576
8.3 Strokes and Aneurysms . . . 577
8.3.1 Hemorrhagic Strokes . . . 578
8.3.2 Ischemic Strokes . . . 587
8.4 Modeling the Circulatory System and the Heart . . . 588
8.4.1 Model of the Heart . . . 588
8.4.2 Model of the Overall Flow in the Circulatory System . . . 592
8.4.3 The Arterial Pulse . . . 596
8.4.4 Windkessel Model . . . 600
8.4.5 Modeling the Malfunctioning Heart . . . 601
8.5 Summary . . . 605
Problems . . . 605
References . . . 618

9 Lungs and Breathing . . . 623
9.1 Structure of the Lungs . . . 624
9.2 The Physics of the Alveoli . . . 628
9.3 Physics of Breathing . . . 632
9.4 Volume of the Lungs . . . 635

9.5 Breathing Under Usual and Unusual Conditions 637
9.5.1 Flow of Air During Breathing 637
9.5.2 Mechanical Model of Breathing and Model Parameters 640
9.5.3 Inspiration/Expiration Cycle 640
9.5.4 Breathing with a Diseased Lung 641
9.5.5 Breathing at Higher Elevations and Under Other Unusual Conditions 644
9.6 Work Needed to Breathe 646
9.7 Summary 647
Problems 648
References 654

10 Sound, Speech, and Hearing 657
10.1 The Physics of Sound Waves 658
10.1.1 The Speed and Properties of Sound Waves 659
10.1.2 Intensity of Sound Waves 661
10.1.3 What Happens When Sound Travels from One Medium to Another? 667
10.1.4 Resonant Cavities 670
10.2 Speech Production 673
10.2.1 Types of Sounds 673
10.2.2 Systems in Speech Production 677
10.2.3 Parameters of the Human Voice 691
10.2.4 The Energetics of Speaking 692
10.3 Hearing 693
10.3.1 Auditory Sensitivity 694
10.3.2 Connections to Hearing Perception 712
10.4 Other Vibrations in the Body 716
10.4.1 Cardiac and Other Sources of Sounds 716
10.5 Summary 719
Problems 719
References 728

11 Light, Eyes and Vision 731
11.1 Structure of the Eye 731
11.2 Focusing and Imaging with Lenses 738
11.2.1 Image Formation 738
11.2.2 Scientific Basis for Imaging 741
11.2.3 Combinations of Lenses or Refractive Surfaces 745
11.3 Imaging and Detection by the Eye 753
11.3.1 Transmission of Light in the Eye 753
11.3.2 The Eye as a Compound Lens 755
11.3.3 Accommodation 761

11.3.4 Field of View and Binocular Vision 763
11.3.5 Adjustments of Light Levels 763
11.3.6 Limitations to Visual Acuity. 765
11.3.7 Imperfect Human Vision 775
11.3.8 Correction of Vision by Eyeglasses, Contact Lenses, and Other Means 780
11.4 Types of Vision Impairment 790
11.5 Connections to Visual Perception 791
11.6 Vision in Other Animals 798
11.7 Summary 802
Problems 802
References 815

12 Electrical and Magnetic Properties 819
12.1 Review of Electrical Properties 820
12.2 Electrical Properties of Body Tissues 824
12.2.1 Electrical Conduction Through Blood and Tissues. . . . 824
12.3 Nerve Conduction 825
12.3.1 Cell Membranes and Ion Distributions. 827
12.3.2 Types of Cell Membrane Excitations. 836
12.3.3 Model of Electrical Conduction Along an Axon 838
12.4 Ion Channels, Hair Cells, Balance, Taste, and Smell 849
12.5 Electrical Properties of the Heart 851
12.6 Electrical Signals in the Brain 859
12.7 Effects of Electric Shock 861
12.8 Magnetic Properties 862
12.8.1 Magnetic Field from an Axon. 863
12.8.2 Magnetic Sense 864
12.9 Electromagnetic Waves 864
12.10 Summary 864
Problems 865
References 870

13 Feedback and Control 873
13.1 Basics of Feedback and Control 874
13.1.1 Control Theory (Advanced Topic). 876
13.2 Regulation of the Body 878
13.2.1 Regulation of Temperature. 878
13.2.2 Control of Blood Pressure 884
13.2.3 Regulation During Exercise 885
13.2.4 Regulation of the Pupil Size. 885
13.3 Summary 887
Problems 887
References 888

Appendix A: Symbols and Units 889

Appendix B: Locator of Major Anatomical and Anthropometric Information 895

Appendix C: Differential Equations 897

Appendix D: Similar Model Systems 905

Appendix E: Biophysics of the Human Body 911

Appendix F: Physics of Life Cycles: Pregnancy, Injuries, Diseases and Disorders, Aging, and Dying 913

Appendix G: Physics of Sports 917

Appendix H: Physics for the Body: Measurements, Imaging, and Treatment 921

Solutions to Selected Problems 927

Index 941